Mohan Krishna

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Education

Birla Institute of Technology and Science (BITS), Pilani Pilani, India
B.E. Mathematics and Computing *Expected May 2028*

- **Relevant Coursework:** Stochastic Calculus, Probability & Statistics, Linear Algebra, Numerical Analysis, Optimization, Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Discrete Mathematics.

Experience

Software Engineering Intern May 2026 – July 2026
OBS Experiences *Dehradun*

- Architected a modular, cloud-native backend in **NestJS (Node.js/TypeScript)** spanning 12+ business domains, including authentication, bookings, payments, trek discovery, loyalty, media, notifications, and user management.
- Shipped two production **Next.js** applications—a customer-facing booking platform and an internal admin dashboard—on a shared component library, cutting duplicate frontend code by **40%+**.
- Automated deployments across four repositories with **GitLab CI/CD, Docker Compose, Nginx, and self-hosted runners**, reducing deployment time by **90%**.
- Authored engineering documentation covering API conventions, ER diagrams, architecture, and development standards, accelerating onboarding and long-term maintainability.

Projects

- IntervieHire – AI Interviewer — LiveKit, WebSockets, Fastify, PostgreSQL Sep 2026

- Migrated real-time AI interviews from Vapi to a **self-hosted LiveKit Agents** pipeline using **Deepgram STT, Silero VAD, Cartesia TTS, and Fastify**, reducing voice infrastructure cost by **58%+** while improving control over real-time interview orchestration.
- Engineered **WebSocket-based bidirectional session orchestration** for low-latency transcript and AI-response streaming, server-authoritative timing, and idempotent interview turns; hardened distributed deployments across Railway/Vercel and migrated interview media storage to **Backblaze B2**.

- official.me Web App Penetration Test Nov 2025 – Dec 2025

- Conducted a web application penetration test focusing on access-control and business-logic vulnerabilities.
- Identified and validated an **Insecure Direct Object Reference (IDOR)** vulnerability enabling authorization bypass on paywalled content and user chats, demonstrated via a controlled proof-of-concept.
- Built a web console and CLI tooling to validate the vulnerability and affected authorization paths.

- OneTap Vendor — Go, Fyne, SQLite, sqlc Dec 2025 – Mar 2026

- Built a desktop billing platform deployed across BITS Pilani eateries, serving **5,000+ students** and supporting **$1.2M+ annual payment volume**.
- Engineered an offline-first synchronization engine with retry and exponential-backoff logic, preventing transaction loss during intermittent connectivity.
- Shipped UPI QR payments, automatic backups, and bill splitting; enforced compile-time type-safe SQL with **sqlc**.

- BOSM (BITS Open Sports Meet) App — React Native, Expo, TypeScript Aug 2025 – Sep 2025

- Developed a cross-platform mobile app used by **6,000+ participants** during BITS Pilani's annual national sports festival.
- Built modules for live match scheduling, automated leaderboard updates, participant verification, and digital food ordering.
- Owned production releases through Expo EAS, Google Play, and the Apple App Store, maintaining stable releases throughout the event.

Leadership

Engineering Lead May 2026 – Present
Students' Union Technical Team, BITS Pilani *Pilani, India*

- Lead a **20-member cross-functional engineering team** across frontend, backend, mobile, and UI/UX, delivering multiple campus products on strict release timelines.
- Introduced structured sprint planning, code reviews, and engineering standards, improving product quality and lifting daily active users across deployed applications by **∼45%**.

Technical Skills

Languages: C++, Python, Go, TypeScript/JavaScript, SQL

Frameworks: Next.js, React, React Native, NestJS, FastAPI, Fastify, Prisma

Databases & Messaging: PostgreSQL, Redis, SQLite

Cloud & DevOps: Docker, Docker Compose, GitLab CI/CD, Nginx, Railway, Vercel, Cloudflare, Backblaze B2, Git, Postman

AI / Realtime: LiveKit Agents, WebSockets, Deepgram STT, Cartesia TTS, Silero VAD

Achievements

- **Top 10 Finalist, Expedia Group Campus Hackathon 2026** — engineered an AI-powered multi-flight journey planner for complex itineraries with constraint-aware route generation and optimized travel recommendations.

Interests

- **Competitive Programming:** LeetCode contest rating 1700+.
- **Trekking & Mountaineering:** completed the week-long Kuari Pass Himalayan trek (12,350 ft).